# Unit 1 Database management systems

Use these model answers to check the corresponding question sheet. Other correct SQL and explanations may satisfy the same requirements.

## 1.1 Compare database users

The lecturer is an end user responsible for accurate authorised entry. The developer implements the interface and queries. The DBA maintains recoverability and performs a tested restore. Permissions should reflect each role.

## 1.2 Plan a database

Create Customers with a stable key, plus Orders with a customer foreign key. Define which phone number is authoritative and resolve duplicates before import. Check customer counts and that each order references an existing customer after migration.

## 1.3 Evaluate costs

Costs include schema and migration work, training, hosting or licences, and administration. A backup must be restorable within the required time; test restores, document procedures and protect backup access.

# Unit 2 Data models and ER diagrams

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## 2.1 Read cardinality

Requester 1 to Ticket 0..many. Ticket.RequesterID is a required foreign key. A requester can exist before creating a ticket, while every ticket must have exactly one existing requester.

## 2.2 Model optional participation

Tickets.AssignedToID is a nullable foreign key to Technicians.TechnicianID. A NULL denotes unassigned. Each ticket has zero or one technician and each technician zero or many tickets.

## 2.3 Choose attribute types

A name can be composite in a conceptual model; date of birth is stored; age is derived for a reference date. Phone numbers are multivalued and belong in PersonPhones with a foreign key, rather than a comma-separated field.

# Unit 3 Relational design and keys

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## 3.1 Choose candidate keys

StudentID and StudentNumber are candidate keys if each is unique and minimal. Choose StudentID as primary and enforce StudentNumber as UNIQUE NOT NULL. Name or City can be search keys but are not reliable row identifiers.

## 3.2 Locate a foreign key

Orders.CustomerID is the foreign key on the many side. It permits many order rows per customer without repeating groups. A customer may have zero orders; each required customer reference must match an existing customer.

## 3.3 Revise an ERD

Introduce EmployeeProjects(EmployeeID PK/FK,ProjectID PK/FK) and remove the single ProjectID from Employees after migration. Add allocation details to the bridge if they depend on that employee-project pair.

# Unit 4 Developing and reviewing ER models

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## 4.1 Unary relationships

Employees has nullable SupervisorID referencing Employees.EmployeeID. One supervisor can supervise many employees. CHECK(SupervisorID<>EmployeeID) prevents direct self-supervision; longer management cycles need additional validation.

## 4.2 Ternary relationships

The fact depends on all three identities together. Use Supply(SupplierID,ProductID,ProjectID,Quantity) with foreign keys and a suitable composite key. Separate pairs cannot prove which supplier delivered which product to a particular project.

## 4.3 Review business rules

Use Titles, Copies, Members and Loans; Copies references Titles and Loans references Copies and Members. A current-loan uniqueness rule needs a filtered unique index or controlled transaction logic, not simply a one-to-many ERD line.

# Unit 5 Normalisation to 3NF

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## 5.1 Partial dependencies

Key is (OrderID,ProductID). OrderID determines OrderDate; ProductID determines ProductName; the whole key determines Quantity. 2NF: Orders(OrderID PK,OrderDate), Products(ProductID PK,ProductName), Lines(OrderID PK/FK,ProductID PK/FK,Quantity).

## 5.2 Transitive dependencies

EmployeeID determines EmployeeName and DepartmentID; DepartmentID determines DepartmentName. Create Departments(DepartmentID PK,DepartmentName) and Employees(EmployeeID PK,EmployeeName,DepartmentID FK). This removes the non-key transitive dependency.

## 5.3 State assumptions

No. It may avoid partial dependency on that selected key, but can still have transitive dependencies, such as EmployeeID -> DepartmentID -> DepartmentName. Other candidate keys must also be considered when determining normal forms.

# Unit 6 SQL commands and queries

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## 6.1 NULL aggregates

COUNT(*)=3, COUNT(Mark)=2, SUM=120 and the mathematical average is 60. SQL Server AVG of integer input returns an integer type; use AVG(CAST(Mark AS DECIMAL(10,2))) when a fractional average is needed.

## 6.2 Filtering groups

SELECT CourseID,AVG(CAST(Mark AS DECIMAL(10,2))) AS AverageMark FROM Enrolments WHERE Mark>=50 GROUP BY CourseID HAVING COUNT(Mark)>=5; WHERE removes failing and NULL rows before groups are counted.

## 6.3 Safe changes

Preview SELECT with the exact StudentID, check enrolments and other dependants, use a controlled practice transaction and DELETE with that predicate, verify affected rows, then commit or roll back. A foreign key may reject deleting a referenced student; do not disable integrity just to force the delete.